Morse Talk: IoT-Based Communication System for Differently Abled People

A PROJECT REPORT
[PHASE II]

Submitted by

KARTHIKEYAN R 113222061009

SUJITH M 113222061026

in partial fulfillment for the award of the degree

of

BACHELOR OF ENGINEERING

IN

ELECTRONICS AND INSTRUMENTATION ENGINEERING

VELAMMAL ENGINEERING COLLEGE, CHENNAI-66

(An autonomous institution, Affiliated to Anna University, Chennai)

APRIL 2026

VELAMMAL ENGINEERING COLLEGE, CHENNAI-66.

BONAFIDE CERTIFICATE

Certified that this project report **"MORSE TALK: IOT-BASED COMMUNICATION SYSTEM FOR DIFFERENTLY ABLE PEOPLE"** is the bonafide work of "**KARTHIKEYAN R (113222061009)** AND **SUJITH M (113222061026)"** who carried out the project work under my supervision.

Dr. Z. BRIJET
PROFESSOR & HEAD
Department of Electronics and Instrumentation Engineering
Velammal Engineering College
Chennai-66.

Mrs. S. PADMA
ASSISTANT PROFESSOR II
Department of Electronics and Instrumentation Engineering
Velammal Engineering College
Chennai-66.

CERTIFICATE OF EVALUATION

COLLEGE NAME : VELAMMAL ENGINEERING COLLEGE

BRANCH : ELECTRONICS AND INSTRUMENTATION ENGINEERING

SEMESTER : VIII

Sl. No	Name of the students	Title of the Project	Name of supervisor with designation
1	KARTHIKEYA R (113220061009)	Morse Talk: IoT-Based Communication System for Differently Abled People	Mrs.S.PADMA Assistant Professor-II
2	SUJITH M (113220061026)		

This report of Project work submitted by the above students in partial fulfillment for theaward of Bachelor of Engineering degree in Electronics and Instrumentation Engineering Degree in Anna University was evaluated and confirmed to be reports of the work done by the above students and then assessed.

Submitted for Internal Evaluation held on.....

Internal Examiner **External Examiner**

ABSTRACT

Assistive technologies have become increasingly vital in enhancing the quality of life for individuals with physical disabilities, particularly those with limited mobility and communication capabilities. This project presents an innovative assistive technology system designed to enable paralyzed individuals to communicate effectively and monitor their well-being. The system integrates an Arduino and ESP32 microcontroller, which operate together to interpret Morse code inputs generated via a simple push button. Each button press is translated into a dot or dash to form letters, which are processed by the Arduino and transmitted to the ESP32. The ESP32 employs the Google Text-to-Speech (TTS) API to convert the text into audible speech, which is amplified through an MP3 decoding board connected to speakers, allowing users to express their thoughts and needs vocally. In addition, the system incorporates a GSM module that enables users to send SMS alerts to caregivers or family members in emergencies. By combining Morse code input, text-to-speech conversion, SMS notifications, and environmental monitoring, the system provides a comprehensive, practical, and efficient tool to enhance the quality of life for individuals with limited mobility, ensuring both effective communication and personal safety

ACKNOWLEDGEMENT

We express our sincere thanks to **Shri. M. V. Muthuramalingam**, Chairman, **Thiru. M. V. M. Velmurugan**, Chief Executive Officer and **Shri. V. Karthick Muthuramalingam**, Deputy Chief Executive Officer for their support during our project work.

We thank **Dr. S. Sathish Kumar**, Principal, Velammal Engineering College for his motivation and support during the course of the project.

We are vey much grateful **to Dr. Z. Brijet**, Professor and Head, Department of Electronics and Instrumentation Engineering, Velammal Engineering College for the encouragement, valuable suggestions and guidance as internal guide during this work.

We are very much indebted to our project guide **Mrs.S. Padma**, Assistant Professor – II, Department of Electronics and Instrumentation Engineering, Velammal Engineering College, for his admirable guidance and constant encouragement throughout the project work.

We would like to extend our fullest gratitude to our Project Coordinator, **D r. S. Anusankari**, Assistant Professor- III and the project review committee for their timely advice and successful grooming.

We take this opportunity to thank our parents, friends, all faculty and staff members of our department for their kind help during the course of the work.

TABLE OF CONTENTS

CHAPTER NUMBER	TOPIC	PAGE NO
	ABSTRACT	**vi**
	ACKNOWLEDGEMENT	**v**
	TABLE OF CONTENT	**vi**
	LIST OF FIGURES	**ix**
	LIST OF ABBREVATIONS	**x**
1	**INTRODUCTION**	
	1.1 GENERAL	1
	1.2 CHALENGES IN EXIXTING SYSTEM	1
	1.3 SCOPE OF THE PROJECT	2
2	**LITERATURE SURVEY**	
	2.1 GENERAL	3
	2.2 RESEARCH WORKS	3
3	**PROPOSED SYSTEM**	
	3.1PROPOSED SYSTEM	8
	3.2 OBJECTIVES	8
	3.3 ARCHITECTURE	9
	3.4 OVERVIEW OF ARCHITECTURE	10
4	**HARDWARE**	
	4.1 HARDWARE COMPONENTS	11
	4.1.1 Esp32	11
	4.1.2 Arduino Uno	12
	4.1.3 Power Supply	13

	4.1.4 Push Button	14
	4.1.5 Oled Display	16
	4.1.6 Speaker	17
	4.1.7 I²S Amplifier	18
	4.1.8 GSM Module	19
5	**SOFTWARE**	
	5.1 SOFTWARE REQUIREMENTS	21
	5.1.1 Arduino Ide	21
	5.1.2 Embedded C	22
6	**DESIGN AND IMPLEMENTATION**	
	6.1 INTRODUCTION	24
	6.2 SYSTEM DESIGN APPROACH	24
	6.3 HARDWARE DESIGN	25
	6.3.1 Input Unit	25
	6.3.2 Processing Unit	25
	6.3.3 Communication Unit	25
	6.3.4 Output Unit	26
	6.4 SOFTWARE DESIGN	26
	6.4.1 Programming Enviroment	28
	6.4.2 Emedded C Implementation	28
	6.5 FLOW OF OPERATION	29
	6.6 IMPLEMENTATION DETAILS	29
	6.7 TESTING AND VALIDATION	30
7	**RESULT AND FUTURE SCOPE**	
	7.1 RESULT	32
	7.2 ANALYSIS OF SYSTEM	32
	7.3 FUTURE SCOPE	33

8 **CONCLUSION**

8.1 CONCLUSION OF THE PROJECT 37

REFERENCES **38**

LIST OF FIGURES

S.NO	FIGURE	PAGE NO
3.1	Block Diagram	9
4.1	ESP32	12
4.2	Arduino UNO	13
4.3	Push Button	15
4.4	OLED Display	16
4.5	Speaker	17
4.6	I^2S amplifier	18
4.7	GSM Module	19
5.1	Arduino IDE	21
5.2	Embedded C	22
7.1	Hardware Output	31
5.2	Software Output	32

LIST OF ABBREVIATIONS

S.NO	TITLE	PAGE NO
1	IoT – Internet of Things	1
2	GSM – Global System for Mobile Communication	7
3	ESP32 – Espressif Systems 32-bit Microcontroller	11
4	OLED – Organic Light Emitting Diode	17
5	I^2S – Inter-IC Sound (audio communication protocol)	21
6	SMS – Short Message Service	26
7	IDE – Integrated Development Environment	30
8	PCB – Printed Circuit Board	30
9	DC – Direct Current	36
10	API – Application Programming Interface	40

CHAPTER 1

INTRODUCTION

1.1 GENERAL

Communication plays a vital role in human interaction, social participation, and daily activities. For individuals with physical or speech disabilities, the lack of effective communication methods creates significant barriers in expressing needs, thoughts, and emotions. Conventional assistive devices often fail to provide reliable or affordable solutions, especially for individuals with severe impairments.

Morse code, a time-tested method of transmitting textual information through sequences of short and long signals, provides a simple yet powerful approach to address this challenge. With the advancement of Internet of Things (IoT) technologies, Morse code can be integrated into digital platforms to create accessible communication systems. Such integration enables disabled individuals to convey messages through minimal physical input, which can then be converted into text or voice outputs and transmitted across connected devices in real time.

This approach not only empowers differently-abled persons with greater independence but also ensures timely communication with caregivers, medical staff, or family members. The combination of Morse code and IoT thus establishes a practical, cost-effective, and inclusive solution in the field of assistive technology..

1.2 CHALLENGES IN EXISTING SYSTEM

Existing communication systems for disabled individuals face multiple limitations. Advanced solutions such as speech-generating devices, brain–computer interfaces, or eye-tracking systems often involve high costs, complex setups.

Moreover, many available devices are not adaptable to different levels of disability. For example, systems relying on fine motor control may not be usable by individuals with severe neuromuscular disorders. Other systems lack portability, making continuous communication outside a controlled environment difficult.

In addition, affordability and availability remain major barriers. High-end assistive devices are beyond the financial reach of many families, while low-cost alternatives often compromise reliability and functionality. These challenges highlight the need for a system that is affordable, adaptable, and easy to use while maintaining accuracy and efficiency in communication.

1.3 SCOPE OF THE PROJECT

The project titled "IoT-Based Communication Using Morse Code for Disabled Persons" aims to design and implement a system that overcomes the shortcomings of existing technologies. The system focuses on translating simple Morse code inputs into meaningful text or speech outputs, which can then be transmitted to connected devices using IoT platforms.

The scope includes the development of a user-friendly interface that requires only minimal physical actions such as pressing a button, blinking an eye, or activating a sensor. The system will incorporate microcontrollers and IoT modules to process the input, generate outputs, and enable communication over short or long distances.

The project also emphasizes cost-effectiveness by relying on affordable hardware components and open-source platforms. Its adaptability allows customization based on the specific needs of different categories of disabilities.

CHAPTER 2

LITERATURE SURVEY

2.1 GENERAL

Literature surveys form an essential part of any research study, as they provide insights into existing knowledge, identify gaps in current systems, and highlight opportunities for further development. In the domain of assistive communication for disabled persons, several researchers have focused on integrating innovative technologies such as IoT, wearable devices, brain–computer interfaces, and alternative communication systems. Morse code, due to its binary simplicity, has also been revisited in multiple studies as a potential method for enabling communication among individuals with severe physical limitations.

This section reviews selected research works relevant to the project "IoT Based Communication Using Morse Code for Disabled Persons". Each work is analyzed in terms of its objectives, contributions, and limitations, with the purpose of building a foundation for the proposed system.

1.Title: A Morse Code-Based Assistive Communication System for Physically Disabled Persons

Author: S. Kumar and A. Mehta

Journal: International Journal of Assistive Technology

Year: 2019

Abstract:

This research proposed a Morse code-based assistive system aimed at individuals with severe motor impairments. The study emphasized the simplicity of Morse code, which can be generated using minimal physical effort such as pressing a switch or blinking an eye. A microcontroller-based device was designed to translate the Morse inputs into alphanumeric text, which was then displayed on a screen. The study highlighted that Morse code requires only two distinct inputs, making it adaptable to a wide range of physical conditions. However, it also noted that the system had limitations in terms of real-time communication and portability. The researchers concluded that while the approach was practical and low-cost, integration with wireless or IoT-based communication platforms would further enhance its usability in daily life scenarios.

2.Title: IoT-Enabled Communication Aid for Speech and Motor Disabled Individuals

Author: R. Patel and M. Singh

Journal: Journal of IoT in Healthcare

Year: 2020

Abstract:

This work introduced an IoT-enabled communication device that converted simple user inputs into text messages, which were transmitted over the internet to caregivers and family members. The device employed basic tactile switches as input mechanisms, allowing users with limited mobility to interact with the system. The outputs were processed using an Arduino microcontroller and displayed through an LCD module. Additionally, the system was connected to an IoT platform to enable long-distance communication. The study emphasized the significance of IoT in enhancing assistive technology by ensuring timely delivery of messages and enabling

remote monitoring. While the system achieved low cost and high accessibility, the researchers observed that user customization was limited.particularly in adapting the input methods for different types of disabilities.

3.Title: Eye-Blink Controlled Morse Code Communication for Paralysis Patients

Author: L. Zhang and Y. Chen

Journal: IEEE Transactions on Biomedical Engineering

Year: 2021

Abstract:

This research explored the application of eye-blink detection as an input method for Morse code-based communication. Using infrared sensors, the system detected voluntary eye blinks and classified them into short and long durations corresponding to Morse code signals. The input was processed in real time and converted into meaningful text using embedded software. The study demonstrated that the system provided an efficient method for patients with paralysis who could not rely on hand or voice inputs. The integration of biomedical sensors enhanced user comfort and reduced the requirement of physical exertion. However, the study pointed out issues such as accidental blinks being misinterpreted as inputs, and the need for improved calibration methods. The authors recommended combining the approach with IoT connectivity for wider communication possibilities.

4.Title: Low-Cost Assistive Devices Using IoT for Differently-Abled Persons

Author: P. Roy and S. Gupta

Journal: International Journal of Emerging Technologies

Year: 2021

Abstract:

This paper discussed the development of low-cost IoT-based assistive devices targeted at differently-abled individuals. The research focused on affordability and accessibility, ensuring that the technology could be adopted even in economically constrained settings. The proposed system included simple input modules, microcontroller processing units, and cloud-based platforms for data transmission. One of the major highlights was the emphasis on modular design, allowing the system to be adapted for various disabilities. The study reported that IoT provided seamless communication between users and caregivers, significantly improving the quality of healthcare and emergency response. Despite the advantages, the authors noted that user training was required to ensure effective operation, and future research should address more intuitive input mechanisms.

5. Title: Assistive Communication through Morse Code Integrated with Wireless Technologies

Author: H. Sharma and V. Nair

Journal: Journal of Communication Systems

Year: 2022

Abstract:

This study investigated the combination of Morse code with wireless communication modules to design a flexible and portable assistive communication system. The system utilized Morse code input through button presses, which were translated into text and transmitted via Bluetooth and Wi-Fi modules. The outputs could be received on smartphones, computers, or dedicated display units. The research highlighted that

the wireless integration eliminated the constraints of traditional wired systems and increased user mobility. Furthermore, the system was designed to be cost-effective, using easily available components. However, the study also identified challenges such as limited range for Bluetooth modules and dependency on stable internet for Wi-Fi communication. Overall, the research concluded that combining Morse code with wireless and IoT technologies offered a practical pathway for developing scalable communication aids for disabled individuals.

CHAPTER – 3

PROPOSED SYSTEM

3.1 Proposed System

The proposed system focuses on developing an IoT-based communication framework that utilizes Morse code as the primary mode of input for disabled individuals. Users can provide inputs through dedicated buttons representing dots, dashes, and emergency signals. These inputs are processed by the Arduino Uno microcontroller, which translates Morse code into corresponding alphanumeric characters. The translated output is displayed on a 16x2 LCD display and simultaneously transmitted through an IoT module (ESP32) for remote communication. Additional components such as a buzzer and DFPlayer Mini module are integrated to provide audio feedback and alert functionalities. The system ensures simple operation, affordability, and real-time communication support, thereby enhancing accessibility for individuals with severe speech or motor disabilities.

3.2 OBJECTIVES OF PROPOSED SYSTEM

1. To design an affordable and accessible communication device for disabled individuals using Morse code as the primary input method.

2. To integrate IoT technology for enabling real-time transmission of translated messages to caregivers or family members.

3. To enhance user interaction through audio feedback, visual display, and emergency alert mechanisms for critical situations.

3.3 ARCHITECTURE

The architecture of the proposed system is designed to ensure smooth interaction between hardware components and efficient data transmission. The block diagram consists of the following elements:

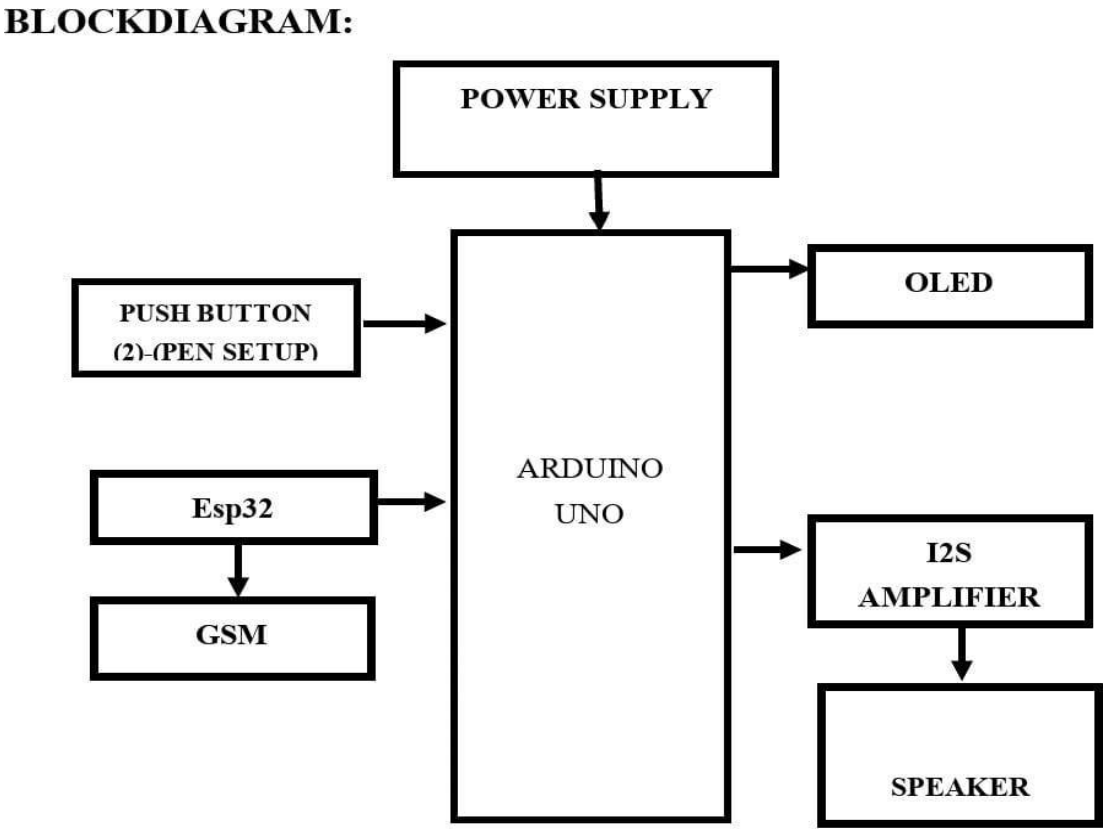

Figure: 3.1 Block Diagram

Input Section: The system includes dedicated buttons for “Dot,” “Dash,” and “Emergency.” These buttons allow the user to provide Morse code inputs or trigger an emergency alert. The simplicity of two inputs makes the system suitable for users with limited physical mobility.

Processing Unit (Arduino Uno): The Arduino Uno serves as the central processing unit. It receives inputs from the buttons, interprets Morse code sequences, and translates them into alphanumeric characters. The processed data is then forwarded to output modules and the IoT interface.

Output Section: The translated characters are displayed on a 16x2 LCD display for immediate feedback. A buzzer is incorporated to provide audio confirmation of inputs, while the DFPlayer Mini module supports audio playback for enhanced interaction.

IoT Communication (ESP32): The ESP32 module handles wireless communication. It transmits the translated messages to remote devices or cloud platforms, allowing caregivers to receive real-time updates. This ensures communication beyond physical proximity, adding significant value in emergency or healthcare scenarios.

Emergency Handling: A separate emergency button is provided to generate priority alerts. This ensures that urgent signals are delivered instantly to caregivers through IoT communication, thereby improving the safety of the user.

3.4 Overview of Architecture:

The architecture demonstrates a seamless integration of input, processing, and output modules with IoT connectivity. The design emphasizes minimal user effort, reliability, and adaptability. By combining Morse code input with communication.

3.5 Advantages of the Proposed System

The proposed Morse Talk communication system provides several benefits when compared with many traditional assistive communication devices. The design focuses on simplicity, affordability, and accessibility so that it can be used by individuals with severe physical disabilities.

One of the main advantages of the system is its simple input mechanism. The communication method relies on Morse code, which consists of only two signals: dots and dashes. Because of this binary input structure, the system can be operated using minimal physical effort such as pressing a button. This makes it suitable for users who may not be able to operate complex keyboards or touch interfaces.

Another major advantage is the low cost of implementation. The device is built using commonly available electronic components such as the Arduino Uno microcontroller, ESP32 module, push buttons, OLED display, and speaker. These components are widely available and relatively inexpensive, which makes the system economically feasible for many users.

The integration of Internet of Things (IoT) technology improves the effectiveness of the communication system. Messages generated by the user can be transmitted instantly to caregivers, family members, or medical staff through wireless networks. This ensures that assistance can be requested quickly when needed.

The system also supports multiple output formats, including visual text displayed on the OLED screen and audio output through a speaker. This dual-output approach improves usability and ensures that messages can be understood easily by others nearby.

Another important feature is the emergency communication capability. By integrating a GSM module, the system can send SMS alerts to preconfigured contact numbers during emergency situations. This provides an additional layer of safety for the user.

Overall, the proposed system provides an effective communication solution that enhances independence and improves the quality of life for individuals with physical or speech impairments.

CHAPTER 4

HARDWARE

4.1 Hardware Requirements

1. ESP 32
2. ARDUINO UNO
3. POWER SUPPLY
4. PUSH BUTTON (2)
5. OLED
6. SPEAKER
7. I2S AMPLIFIER
8. GSM

4.1.1 ESP32 (IoT Module)

Features:

- Dual-core microcontroller with clock speeds up to 240 MHz.
- Integrated Wi-Fi (802.11 b/g/n) and Bluetooth Classic / BLE.
- Multiple communication peripherals: UART, SPI, I^2C, I^2S, ADC, DAC, PWM.
- Native support for low-power modes (deep sleep) for energy-efficient operation.
- Operates at 3.3 V logic and includes flash memory for firmware.

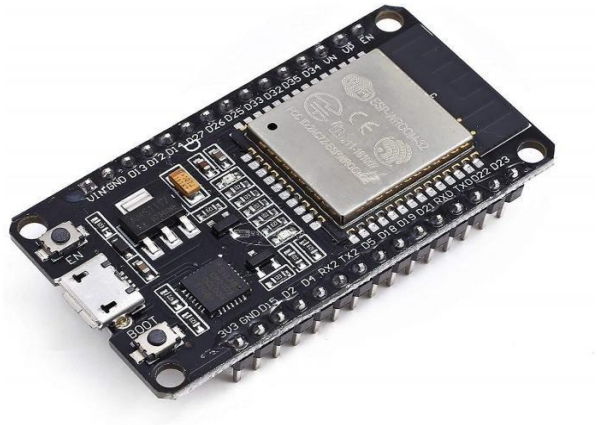

Figure: 4.1 EPS32

Working:

The ESP32 functions as the wireless communication and processing interface for cloud connectivity, remote notifications, and audio streaming via I²S. Wireless data transmission occurs over Wi-Fi or BLE using TCP/IP stacks or MQTT protocols. UART or SPI provides serial data exchange with the primary microcontroller (Arduino Uno) for message forwarding or command control. When audio playback from cloud or local files is required, the ESP32 can produce an I²S digital audio stream for the I²S amplifier. Power management features enable suspension of radio modules to conserve energy during idle periods.

4.1.2 Arduino Uno

Features:

- Based on the ATmega328P microcontroller with a clock speed of 16 MHz.
- Equipped with 14 digital input/output pins (6 PWM-capable) and 6 analog input pins.
- Operates on a 5V regulated power supply, with an onboard regulator for stable performance.
- Includes 32 KB flash memory, 2 KB SRAM, and 1 KB EEPROM for storage.
- Provides multiple communication interfaces such as UART, SPI, and I²C.

Figure: 4.2 Arduino UNO

Working:

The Arduino Uno acts as the central controller in the Morse code communication project. It receives input signals from the push buttons and processes them to differentiate between dots and dashes. The microcontroller then decodes these inputs into corresponding characters using predefined lookup tables in the Embedded C program. The processed data is displayed on the OLED and transmitted to the ESP32 or GSM module for further communication. The Uno also controls the speaker and amplifier to provide audio feedback. With its versatile I/O support and real-time processing capability, the Arduino Uno ensures reliable operation of the entire system.

4.1.3 Power Supply

Features:

- Regulated DC outputs commonly required: 5 V and 3.3 V rails.
- Adequate current capability; recommended 2 A to 3 A minimum for systems

including GSM or audio amplifiers.

- Protection features such as over-current, short-circuit, and over-voltage protection.
- Option for switching (buck) regulators for high efficiency or linear regulators for simple low-noise applications.
- Input options: DC adapter, battery pack, or USB power source.

Working:

A regulated power supply provides stable voltage rails for all system modules. The 5 V rail powers the Arduino Uno and peripherals that require 5 V logic. The 3.3 V rail powers ESP32, I2S amplifier, and other 3.3 V devices. Decoupling capacitors placed close to module power pins smooth transient currents. In designs including GSM modules, the power subsystem must supply short high-current bursts during transmission.

4.1.4 Push Button (2) — Dot and Dash Inputs

Features :

- Momentary tactile switches rated for low-voltage DC logic use.
- Typical mechanical life >100,000 actuations.
- Small footprint for panel or breadboard mounting.
- Operates reliably with pull-up or pull-down resistor configurations.

Working :

Two push buttons act as dedicated input keys: one for the dot symbol and one for the dash symbol in Morse code. Button signals connect to digital input pins on the microcontroller using either internal or external pull-up/pull-down resistors. Button

press events are debounced in firmware, either by sampling over a short interval or via hardware RC filtering. Press-duration logic or separate keys distinguish dots from dashes. Press sequences are time-stamped to

Figure: 4.3 Push Button

determine inter-character and inter-word gaps so that correct mapping from sequence to alphanumeric character can occur.

Additional firmware logic monitors input timing to ensure accurate symbol recognition. The decoded characters are then displayed on an LCD or sent via serial communication for verification. A visual LED indicator or buzzer provides feedback for each valid keypress. The system can also support training modes to help users learn Morse code entry. Power-efficient design techniques minimize current consumption during idle states.

4.1.5 OLED Display

Features:

- High-contrast, low-power OLED panel, typical resolutions include 128×64 pixels.
- Driver commonly based on SSD1306 or equivalent.
- Interfaces available: I²C (two-wire) or SPI (four-wire).
- Wide viewing angles and fast response time.
- Minimal backlight requirement; suitable for battery-powered systems.

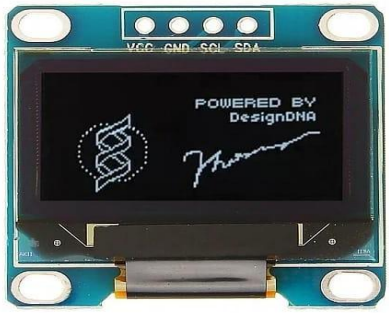

Figure: 4.4 OLED Display

Working:

The OLED display provides visual feedback of decoded characters, system status, and menus. Communication with the microcontroller occurs over I²C or SPI, allowing simple wiring and low pin usage. Display updates are performed by sending frame buffer commands from the controller; partial updates are possible to minimize latency. Brightness and contrast are adjustable via firmware parameters. The low-

power nature of OLED technology reduces overall system consumption when combined with sleep modes.

4.1.6 Speaker

Features:

- Speaker impedance and power rating selected to match amplifier output (commonly 4 Ω or 8 Ω, small signal types for embedded use).
- Compact form factors available for integration into enclosures.
- Frequency response sufficient for voice or tone playback in assistive applications.

Figure: 4.5 Speaker

Working:

The speaker converts amplified electrical audio signals into audible sound for user prompts, Morse feedback tones, and voice messages. The audio signal received from the I^2S amplifier drives the speaker across an output stage. For clear speech playback, the speaker should be paired with an amplifier capable of delivering adequate power

and low distortion. Enclosure mounting and acoustic coupling influence perceived loudness and intelligibility.

4.1.7 I2S Amplifier

Features:

- Digital audio amplifier module utilizing I²S input (example class: MAX98357A-style modules).
- Class-D operation for high efficiency and low heat dissipation.
- Direct interface to microcontroller digital audio output (ESP32 supports native I²S).
- Compact size and low-power operation.

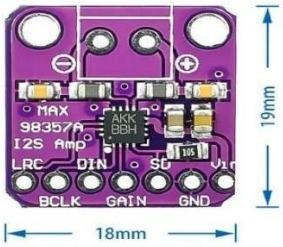

Figure: 4.6 I2S Amplifier

Working:

The I²S amplifier receives a digital audio stream from the ESP32 via the I²S bus and performs digital-to-analog conversion followed by amplification suitable for driving

the speaker. Digital audio format and sample rate (e.g., 16 kHz, 44.1 kHz) must be configured on the source. Because amplification occurs in the digital domain until final conversion, audio quality remains high and cable noise susceptibility is reduced. Volume control may be implemented in software by scaling digital samples or via amplifier register controls where supported.

4.1.8 GSM Module

Features:

- Cellular modem supporting 2G/3G/4G (module model dependent; common modules: SIM800, SIM900, or modern LTE variants).
- Supports SMS, voice calls, and GPRS/HTTP/TCP data services.
- UART AT-command interface for control and network operations.
- Requires high current capability during transmission bursts; typical peak current up to 2A

Figure: 4.7 GSM Module

Working:

The GSM module provides network-level communication over cellular infrastructure for SMS alerts, voice calls, or data upload where Wi-Fi coverage is unavailable.

Control occurs via AT commands sent over UART from the microcontroller. Typical workflows include SIM initialization, network registration, SMS composition and send, or establishment of TCP connections for data upload. Power supply design must account for transient current draw during transmission; series protection and sufficient decoupling capacitors are mandatory. Antenna placement and signal strength influence reliability.

4.2 Hardware Integration

Hardware integration plays an important role in ensuring that all components of the Morse Talk system function together efficiently. The system combines multiple electronic modules that communicate with each other to perform input processing, message decoding, and communication tasks.

At the center of the hardware setup is the Arduino Uno microcontroller, which acts as the primary control unit. The push buttons used for Morse code input are connected to the digital input pins of the Arduino. These buttons allow users to generate dot and dash signals that form the Morse code characters.

The OLED display module **is** connected to the Arduino through the I²C communication protocol. This interface uses only two communication lines, which simplifies wiring and reduces the number of required connections. The display shows the decoded text messages and system status information.

The ESP32 module is integrated into the system to provide wireless communication capabilities. It communicates with the Arduino through serial UART communication. The ESP32 allows the system to connect to Wi-Fi networks and transmit messages over the internet.

In addition, a GSM module is included to provide cellular communication support. This module enables the system to send SMS alerts to caregivers or emergency contacts when necessary. The Arduino sends AT commands to the GSM module to initiate message transmission.

The I^2S amplifier and speaker are used to generate audio output. The amplifier receives digital audio signals and converts them into amplified analog signals that drive the speaker. This allows the system to produce audible feedback or speech output.

All hardware components are powered using a regulated power supply to maintain stable voltage levels. Proper grounding and careful wiring are necessary to ensure reliable system operation and prevent electrical interference between modules.

CHAPTER 5

SOFTWARE

5.1 Software Requirements

5.1.1 Arduino IDE (Integrated Development Environment)

Features :

- Cross-platform application for writing, compiling, and uploading firmware to Arduino-compatible boards.
- Built-in board manager for selecting target microcontroller and cores.
- Extensive library ecosystem for peripheral drivers (SSD1306, Wi-Fi, I²S, GSM, etc.).

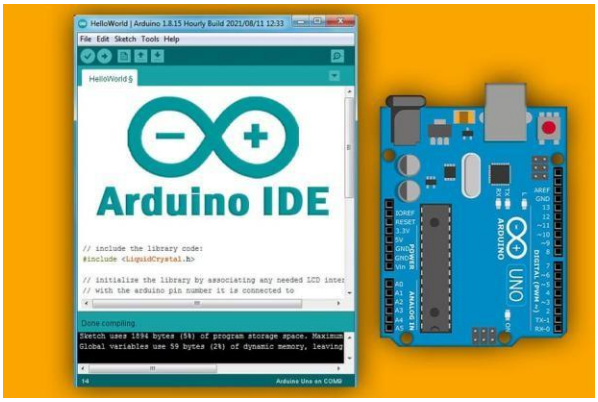

Figure: 5.1 Arduino IDE

Working :

The Arduino IDE compiles program code using a toolchain (AVR-GCC for AVR-based boards) and uploads the resulting binary to the Arduino Uno over USB. Libraries simplify peripheral access, providing functions for digital I/O, timers, serial

communication, and display control. The serial monitor allows observation of runtime messages, aiding debugging and calibration. For ESP32 development, the Arduino IDE can host ESP32 cores to compile and flash firmware directly.

5.1.2 Embedded C Language

Features :

- Low-level programming language enabling direct hardware access and efficient.
- Support for interrupts, precise timer management, and direct register manipulation.
- Small runtime footprint suitable for microcontrollers with limited resources.
- Portable across microcontroller families with appropriate toolchain support.

Figure: 5.2 Embedded C

Working :

Embedded C serves as the primary language for firmware development on both the Arduino Uno and, if required, other microcontrollers. Key functionalities implemented in embedded C include button debouncing logic, timing measurement for dot/dash classification, Morse code mapping tables, serial communication handlers for ESP32 and GSM modules, display update routines, and audio control commands. Interrupt Service Routines (ISRs) manage time-critical events such as

precise interval measurement, while main loop code handles higher-level state transitions and communication tasks.

Embedded C is the core programming language used in the Morse code communication project, enabling smooth interaction between hardware components and ensuring reliable execution. The push buttons are programmed as digital inputs, and their signals are detected, debounced, and stored as Morse code sequences. These sequences are then translated into corresponding alphanumeric characters using lookup tables within the program. The processed output is displayed on the OLED display, while the speaker and amplifier provide audio feedback for user interaction. Embedded C also manages IoT and GSM modules, allowing the translated messages to be transmitted wirelessly to caregivers in real time. Its efficient, lightweight structure ensures fast response, reliability, and ease of debugging for the system.

5.2 Software Architecture

The software architecture of the Morse Talk communication system is designed to ensure smooth interaction between hardware components while maintaining efficient processing of user inputs. The software is structured in a modular manner so that each functional unit of the system performs a specific task.

The first module in the software structure is the input monitoring module. This module continuously checks the status of the push buttons connected to the Arduino. When a button press is detected, the system records the duration of the press to determine whether it represents a dot or a dash in Morse code.

The next module is the Morse code decoding module**.** This part of the program processes the sequence of dots and dashes entered by the user. The input sequence is compared with predefined Morse code patterns stored in the program memory. Based on this comparison, the system determines the corresponding alphabet, number, or symbol.

Once a character has been successfully decoded, the output processing module is activated. This module sends the decoded text to the OLED display for visual representation. At the same time, the system may generate audio signals through the speaker to provide feedback for the user.

The communication module is responsible for transmitting the decoded message to external devices. The ESP32 module is used for Wi-Fi based communication, allowing the message to be sent through internet networks. In emergency situations, the GSM module can send the message as an SMS alert to predefined contacts.

This modular software design simplifies system maintenance and allows future improvements such as mobile application support, predictive text features, and integration with cloud platforms.

CHAPTER – 6

DESIGN AND IMPLEMENTATION

6.1 Introduction

The design and implementation phase represents the translation of theoretical concepts into a working prototype. The IoT-based Morse code communication system for disabled persons is structured into hardware and software modules that work in synchronization to achieve the project objectives. The design methodology ensures that the proposed system is modular, scalable, and capable of real-time communication. This chapter explains the step-by-step design, hardware integration, software development, and implementation strategies adopted to realize the system.Monitoring Parameters:

6.2 System Design Approach

The project design is divided into three stages:

1. Input Stage: Responsible for capturing user actions through push buttons.
2. Processing Stage: Managed by the Arduino Uno, which converts button presses into Morse code and decodes them into characters.
3. Communication and Output Stage: Handled by the ESP32, GSM module, OLED, and speaker to provide visual, audio, and remote communication outputs.

This modular approach ensures flexibility in design and simplifies troubleshooting, as each stage is independently tested before integration.

6.3 Hardware Design

The hardware design integrates multiple components that function together to create a reliable communication system.

6.3.1 Input Unit

Two push buttons are provided for Morse code entry, where one represents a dot and the other a dash. An additional button can be integrated for emergency alerts. These inputs are fed to the Arduino Uno through its digital pins.

6.3.2 Processing Unit

The Arduino Uno acts as the primary controller, executing the Embedded C program for decoding Morse inputs. It differentiates between dot, dash, and inter-character spacing based on button press durations.

6.3.3 Communication Unit

- The ESP32 module enables wireless communication through Wi-Fi or Bluetooth for IoT-based applications.
- The GSM module supports cellular communication for SMS alerts in emergencies.
- Both modules work in coordination with the Arduino Uno via UART communication.

6.3.4 Output Unit

- An OLED display provides real-time visual feedback of decoded characters.
- A speaker, driven through an I2S amplifier, produces audible signals and voice messages to assist visually impaired users.
- The buzzer or speaker can also be programmed to provide input confirmation tones.well-being.

6.4 Software Design

The software design of the Morse Talk: IoT-Based Communication for Disabled People project focuses on creating a reliable, user-friendly, and efficient communication system that translates Morse code inputs into text or voice output. The design follows a modular structure, ensuring that each component performs a specific function while maintaining overall system integrity and scalability.

At the core of the system lies an Arduino/ESP32 microcontroller, which acts as the main processing unit. The software is designed to capture input signals from the user—either through push buttons or touch sensors—and process them as Morse code sequences. Each short press is interpreted as a “dot” and each long press as a “dash.” The program then uses conditional loops and timing functions to differentiate between these signals.

A lookup table algorithm is implemented to match each Morse code pattern to its corresponding alphabet or number. Once the decoding is complete, the translated text is displayed on an LCD or OLED screen, and optionally, it can be converted into speech using a text-to-speech module.

To enhance the user experience, the design also includes debouncing logic to prevent false readings from button noise and a real-time delay adjustment system to accommodate users with different typing speeds. The system can further connect to an IoT platform to send or receive messages over Wi-Fi.

This structured design allows for easy future upgrades such as integrating cloud storage, predictive text algorithms, and AI-based word suggestions. Overall, the software design ensures reliability, modularity, and simplicity, making it accessible for people with disabilities while maintaining efficient system performance.

The Morse Talk system was developed in a flexible and open-source programming environment suitable for IoT and embedded system applications. The primary development platform used is the Arduino Integrated Development Environment (IDE), which provides a simple interface for writing, compiling, and uploading code to the microcontroller.

The programming language used is Embedded C/C++, known for its efficiency and direct hardware control. The Arduino IDE was chosen because it supports a wide range of microcontrollers (such as Arduino Uno, ESP8266, and ESP32) and offers a vast library ecosystem for sensors, displays, and communication modules.

The ESP32 (or similar IoT-enabled board) acts as the main controller due to its built-in Wi-Fi and Bluetooth features. Libraries such as WiFi.h, HTTP Client.h, and Liquid Crystal.h are used to handle communication, display, and input functions. The Morse decoding logic is implemented through timing-based conditions and arrays, ensuring accurate detection of dots, dashes, and word spaces.

For data transmission and IoT functionality, the device can connect to a cloud platform like ThingSpeak, Blynk, or Firebase. This allows real-time communication and future expansion toward storing user data or sending alerts through the internet.

Testing and debugging were performed within the Arduino IDE serial monitor, where input and output signals could be monitored and adjusted for accurate performance. The environment also supports integration with external applications such as the MIT App Inventor or Blynk mobile app, enabling remote message display or control through smartphones.

6.4.1 Programming Environment

The Arduino IDE is used to write, compile, and upload the Embedded C program to the Arduino Uno and ESP32. Standard libraries such as Wire (for I2C), Adafruit SSD1306 (for OLED), and GSM libraries are utilized.

6.4.2 Embedded C Implementation

The software structure is divided into the following segments:

1. Initialization: All input pins, output pins, and communication ports are configured.
2. Input Reading: Button presses are continuously monitored, and debounce routines ensure clean signal detection.
3. Morse Decoding: Timers and lookup tables map button press sequences into dots, dashes, and subsequently into characters.
4. Output Display: Decoded text is displayed on the OLED and transmitted to GSM/ESP32 for communication.

5. Audio Feedback: The speaker produces tones or voice messages through the I2S amplifier.

6.5 Flow of Operation

The operation of the system follows a structured sequence

1. User Input: The disabled person presses the dot or dash button to form a Morse code sequence.
2. Signal Capture: Arduino Uno reads the digital input and stores the sequence temporarily.
3. Decoding: The Embedded C program matches the sequence with predefined Morse code characters.
4. Display and Audio: The decoded character is shown on the OLED and optionally announced through the speaker.
5. Communication: The complete message is transmitted via ESP32 (IoT) or GSM (SMS) to caregivers.
6. Feedback and Reset: The system provides feedback to confirm transmission and resets for the next input.

6.6 Implementation Details

The implementation process began with the integration of the input hardware, where push buttons were connected to the digital pins of the Arduino Uno. A power supply unit was configured to provide stable voltages to the Arduino, ESP32, and GSM modules, each requiring specific power levels. Once the hardware was successfully connected, the software program was developed and uploaded through the Arduino IDE. The coding process was iterative, involving continuous testing to ensure that

button press timings were accurately detected and correctly mapped into Morse code symbols. Particular focus was given to handling timing intervals, as accurate differentiation between dots, dashes, and spacing is essential for correct decoding. The integration of the OLED ensured that real-time feedback was visible to users, while the I^2S amplifier allowed the system to produce clear audio outputs. The GSM module required additional attention during implementation due to its higher current demands and network initialization times. Once configured, it successfully transmitted SMS alerts to registered numbers. Similarly, the ESP32 was programmed for IoT communication, enabling the transmission of decoded messages over Wi-Fi networks. The final implementation ensured smooth synchronization between input, processing, and output modules.

6.7 Testing and Validation

The system was tested in various stages to ensure its robustness. During the hardware testing phase, each component was verified individually to confirm proper functionality. The push buttons were tested for responsiveness and reliability, while the OLED was checked for correct character display. The GSM and ESP32 modules were validated by sending test messages through both SMS and Wi-Fi channels. In the software testing phase, Morse code sequences were manually entered to evaluate decoding accuracy. The system consistently provided correct translations, indicating that the timing logic was functioning as expected. Once the unit testing was complete, integration testing confirmed that all modules worked together seamlessly.

CHAPTER 7
RESULTS AND FUTURE SCOPE

7.1 Results of the Proposed System

The prototype system was developed and tested under various conditions to evaluate its functionality. The outcomes confirm that the design objectives have been successfully achieved.The system consistently identified dots and dashes based on button press durations, achieving a high level of decoding accuracy. The OLED display provided real-time feedback by displaying decoded characters with minimal delay.

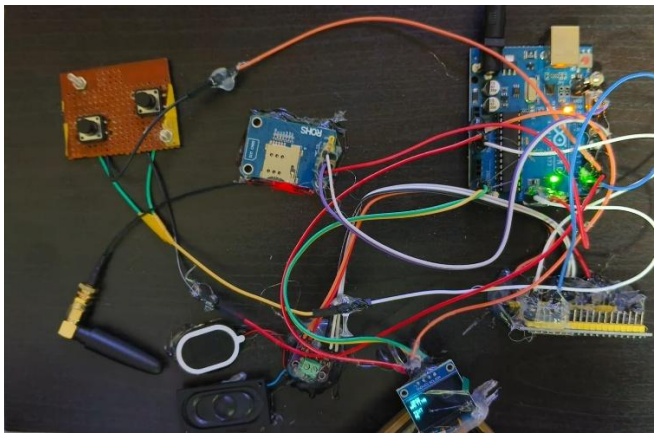

Figure: 7.1 Hardware Output

The speaker generated audible confirmation tones, which enhanced accessibility and usability for users.Communication through ESP32 and GSM modules was also tested successfully. The ESP32 facilitated message transfer over Wi-Fi networks, while the GSM module enabled SMS alerts to predefined numbers. This ensured that messages

could be delivered reliably even in the absence of internet connectivity. The combination of visual, audio, and mobile outputs made the system easy to operate and adaptable to different user needs.

7.2 Analysis of System Performance

The integration of hardware and software components resulted in stable and efficient system performance. Morse code decoding achieved a high success rate during practical trials, while the communication modules transmitted information accurately across both IoT and GSM channels.

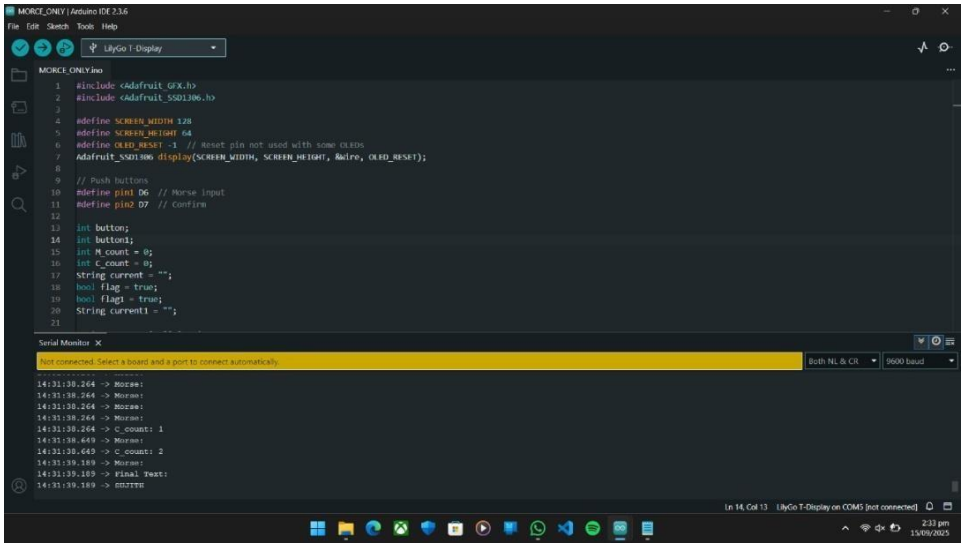

Figure: 7.2 Software Output

The OLED display offered clear and legible text output, and the speaker produced audible signals at an appropriate volume. The system also demonstrated low power consumption, ensuring continuous usage without excessive energy demand.

7.3 Future Scope

The project has considerable potential for further development and practical deployment. A mobile application can be introduced to interface with the system through Bluetooth or Wi-Fi, allowing more convenient communication. Voice-to-text and text-to-voice capabilities may be integrated to support users requiring auditory or verbal communication .The incorporation of machine learning techniques could enable predictive typing, reducing the time and effort required to compose full messages. Miniaturizing the system into a compact or wearable format, such as a wristband or glove, would increase portability and comfort. Additionally, support for multiple languages could extend its use to a broader population worldwide, thereby enhancing inclusivity.

The Morse Talk: IoT-Based Communication for Disabled People project has significant potential for future development and real-world applications. Although the present prototype successfully enables communication using Morse code input through push buttons and IoT integration, there is wide scope for enhancement in terms of user interface, data processing, cloud connectivity, and intelligent automation. The following points outline the possible directions for future improvement and expansion.

In the current design, a push button or a simple tactile input device is used to generate Morse code signals. While this approach is functional, it may not be convenient for all users, especially those with severe physical disabilities or limited motor control. In the future, the system can be upgraded by integrating a touch display interface or a capacitive touch panel to replace or supplement the push button.

A touch display would allow users to interact more intuitively with the system through virtual buttons or customizable on-screen keys. The interface could display visual feedback for every input, such as short and long presses representing dots and dashes, helping users to confirm their entries in real time. Moreover, the touch interface can be designed with adjustable sensitivity and large-sized icons to suit the specific needs of individuals with varying degrees of physical ability.

Another promising area for future enhancement is the integration of cloud technology. By connecting the system to cloud servers, the device can store frequently used words, sentences, or Morse patterns in a personal database. This would enable faster communication by predicting or auto-suggesting words that the user types frequently.

For example, if the user often inputs words like “help,” “food,” or “water,” the system can learn from their usage patterns and display these as quick-access options on the screen. Cloud storage also ensures that the user’s personalized settings and data are not lost even if the device is reset or replaced.

Furthermore, cloud integration can facilitate remote monitoring and analytics, allowing caregivers, doctors, or family members to track the communication history, response time, and behavioral patterns of the user. This information can be highly valuable in healthcare environments, where understanding the communication habits of differently abled individuals helps in providing better care and support.

By leveraging cloud connectivity, Morse Talk can be extended to synchronize data across multiple devices in real time. A user’s profile can be accessed from various platforms such as smartphones, tablets, or assistive communication devices. This

feature would make the system more flexible and adaptable to different environments—for instance, the user could start a conversation on a personal device and continue it seamlessly on another without losing data or progress.

Such synchronization also allows multiple caregivers or family members to receive real-time notifications or messages from the user, ensuring timely assistance in case of emergencies.

Future versions of the project can include AI-based features for voice synthesis, gesture recognition, and predictive text generation. Machine learning algorithms can analyze the user's past communication patterns and predict complete sentences from partial Morse inputs, thereby reducing typing effort and time. AI-driven voice output can also make communication more natural and expressive, improving the overall user experience.

Another area of future work involves optimizing the system for low power consumption and compact design. By using efficient microcontrollers and low-energy wireless modules (such as ESP32 or BLE-enabled boards), the entire system can be made portable and battery-operated. Integration with wearable devices, such as smartwatches or headbands with touch or blink sensors, can make communication even more effortless for users with limited mobility.

As the system grows to include cloud-based storage and IoT connectivity, ensuring data security becomes essential. Future improvements could incorporate encryption protocols and secure authentication mechanisms to protect user data and maintain privacy, especially when medical or personal information is transmitted over the internet.

In the future, the Morse Talk system can be extended to work seamlessly with Android and iOS mobile applications. This would allow users to operate the device through their smartphones or tablets, increasing accessibility and portability. A dedicated mobile app can provide features such as real-time message translation, customizable input speed, and language selection. It can also support voice output using the phone's built-in text-to-speech engine.

Additionally, the mobile interface can sync with cloud storage to maintain a record of frequently used words, emergency contacts, and previously sent messages. This integration will enable users to communicate effortlessly both indoors and outdoors without needing a separate hardware setup.

7.4 Limitations of the Current System

Although the developed Morse Talk communication system demonstrates effective functionality, certain limitations exist in its current implementation. Identifying these limitations is important for guiding future improvements and enhancements.

One limitation of the system is the **requirement for users to learn Morse code**. While Morse code uses only two basic signals, users must still become familiar with the patterns that represent different letters and numbers. This may require training and practice before the system can be used efficiently.

Another limitation is related to the **physical input method**. The current system uses push buttons to generate Morse signals. For individuals with extremely limited hand movement, operating these buttons may still be challenging.

The communication capabilities of the system also depend on **network availability**. IoT-based message transmission requires a stable Wi-Fi connection, and GSM communication

relies on cellular network coverage. In areas with poor connectivity, message transmission may be delayed.

In addition, the current prototype consists of several individual modules connected through external wiring. This increases the physical size of the system and reduces its portability. A more compact design would improve usability and convenience.

Despite these limitations, the system provides a strong foundation for assistive communication technology. Future developments such as touch interfaces, wearable sensors, mobile applications, and intelligent prediction algorithms can further enhance the system’s performance and accessibility

CHAPTER –8

CONCLUSION

8.1 Conclusion:

The IoT-based Morse code communication system successfully demonstrates how assistive technologies can bridge communication gaps for individuals with severe physical limitations. By combining the simplicity of Morse code input with modern microcontrollers such as Arduino Uno and ESP32, and integrating multi-channel outputs like OLED displays, GSM messaging, and audio feedback, the system delivers a reliable, user-friendly, and cost-effective solution. Its modular hardware and software design ensures that it can be customized for different needs while maintaining high accuracy in decoding and message transmission.